

Session 09 (AIML) -Assignment Question

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- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML

Q1.

PROGRAM:

```
session 9.py > ...
1  # a) Empty dictionary using two methods
2
3  d1 = {}          # Method 1
4  d2 = dict()      # Method 2
5
6  print(d1, type(d1))
7  print(d2, type(d2))
8
9  # b) Dictionary with string keys
10 student = {
11     "name": "Shruti",
12     "city": "Pune",
13     "course": "Python"
14 }
15
16 print(student)
17 print(type(student))
18
19 # c) Dictionary with integer keys
20 marks = {
21     1: 90,
22     2: 85,
23     3: 95
24 }
25
26 print(marks)
27 print(type(marks))
28
29 # d) Mixed data type dictionary
30 mixed = {
31     "name": "Shruti",
32     "age": 20,
33     "percentage": 89.5
34 }
35
36 print(mixed)
37 print(type(mixed))
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> & 'C:\Users\tanve\AppData\Local\Python
-x64\bundled\libs\debugpy\launcher' '59994' '--' 'c:\Users\tanve\Desktop
{} <class 'dict'>
{} <class 'dict'>
{'name': 'Shruti', 'city': 'Pune', 'course': 'Python'}
<class 'dict'>
{1: 90, 2: 85, 3: 95}
<class 'dict'>
{'name': 'Shruti', 'age': 20, 'percentage': 89.5}
<class 'dict'>
PS C:\Users\tanve\Desktop\ITR>
```

Q2.

PROGRAM:

```
session 9.py > ...
1  student = {
2      "name": "Shruti",
3      "age": 20,
4      "city": "Pune",
5      "marks": 85
6  }
7
8  # Print name using square brackets
9  print(student["name"])
10
11 # Update marks
12 student["marks"] = 92
13
14 # Add new key
15 student["grade"] = "A"
16
17 # Print updated dictionary
18 print(student)
```

```
PS C:\Users\tanve\Desktop\ITR> c:: cd 'c:\Users\tanve\Desktop\ITR'; & 'C:\Users\tanve\AppData\Local\Microsoft\WindowsApps\python.exe' -x c:\Users\tanve\AppData\Local\Microsoft\WindowsApps\python.exe --help
Shruti
{'name': 'Shruti', 'age': 20, 'city': 'Pune', 'marks': 92, 'grade': 'A'}
PS C:\Users\tanve\Desktop\ITR>
```

PROGRAM:

21 pr

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\Users\tanve\Desktop\ITR'; & 'C:\Users\tanve\AppData\Local\Python\python\python.exe' -i -c 'import sys; sys.path.append('C:\Users\tanve\AppData\Local\Python\python\python.exe'); import os; os.chdir('C:\Users\tanve\Desktop\ITR'); import json; data = {'name': 'shruti', 'age': 21, 'profession': 'Engineer'}; print('Age =', data['age']); print('Salary =', data.get('salary', 'Not Available')); print('Keys:', dict_keys(data.keys())); print('Values:', dict_values(data.values())); print('Items:', dict_items(data.items()));'
```

Q4.

PROGRAM:

```
session 9.py > ...
1  students = {
2      "s1": {"name": "Rahul", "age": 20, "marks": 88},
3      "s2": {"name": "Sneha", "age": 21, "marks": 95}
4  }
5
6  # First student details
7  print(students["s1"])
8
9  # Marks of second student
10 print("Second student marks =", students["s2"]["marks"])
11
12 # Add math marks to first student
13 students["s1"]["math"] = 90
14
15 print(students)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\Users\tanve\Desktop\ITR'; & 'C:\Users\tanve\AppData\Local\Python\python\python.exe' -i -c 'import sys; sys.path.append('C:\Users\tanve\AppData\Local\Python\python\python.exe'); import os; os.chdir('C:\Users\tanve\Desktop\ITR'); import json; data = {'name': 'shruti', 'age': 21, 'profession': 'Engineer'}; print('Age =', data['age']); print('Salary =', data.get('salary', 'Not Available')); print('Keys:', dict_keys(data.keys())); print('Values:', dict_values(data.values())); print('Items:', dict_items(data.items()));'
```

Q5.

PROGRAM:

```
session 9.py > ...
1  mobile = {
2      "brand": "Samsung",
3      "model": "A52",
4      "price": 25000
5  }
6
7  # Update dictionary
8  mobile.update({
9      "color": "Black",
10     "price": 24000
11 })
12
13 # Remove model
14 removed = mobile.pop("model")
15 print("Removed value =", removed)
16
17 # Final dictionary
18 print(mobile)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\Desktop\ITR'
Removed value = A52
{'brand': 'Samsung', 'price': 24000, 'color': 'Black'}
PS C:\Users\tanve\Desktop\ITR>
```

Q6.

PROGRAM:

```
session 9.py > ...
1  # pop() removes a specific key.
2  # popitem() removes the last inserted key-value pair.
3
4  data = {
5      "a": 10,
6      "b": 20,
7      "c": 30,
8      "d": 40,
9      "e": 50
10 }
11
12 # popitem() first time
13 item1 = data.popitem()
14 print("Removed:", item1)
15
16 # popitem() second time
17 item2 = data.popitem()
18 print("Removed:", item2)
19
20 # Clear dictionary
21 data.clear()
22
23 print("After clear:", data)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\Users\tanve\
ions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\
Removed: ('e', 50)
Removed: ('d', 40)
After clear: {}
PS C:\Users\tanve\Desktop\ITR> 
```

Q7.

PROGRAM:

```
session 9.py > ...
1  d = {"a": 1, "b": 2}
2
3  # Copy dictionary
4  copy_d = d.copy()
5
6  # Add key c
7  d.setdefault("c", 3)
8
9  # Existing key a
10 d.setdefault("a", 100)
11
12 print("Original =", d)
13 print("Copied =", copy_d)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:
ions\ms-python.debugpy-2026.6.0-win32-x64\
Original = {'a': 1, 'b': 2, 'c': 3}
Copied = {'a': 1, 'b': 2}
PS C:\Users\tanve\Desktop\ITR> 
```

Q8.

PROGRAM:

```
session 9.py > ...
1  keys = ["name", "age", "city"]
2
3  # Create dictionary
4  d = dict.fromkeys(keys)
5
6  # Input values
7  for key in d:
8      d[key] = input("Enter " + key + ": ")
9
10 print(d)
11
12 # Check key
13 k = input("Enter key to search: ")
14
15 if k in d:
16     print("Key exists")
17 else:
18     print("Key does not exist")
```

OUTPUT:

```
copied - {0: 1, 0: 2}
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\Desktop\ITR
ions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\lau
Enter name: shruti
Enter age: 18
Enter city: pune
{'name': 'shruti', 'age': '18', 'city': 'pune'}
Enter key to search: age
Key exists
PS C:\Users\tanve\Desktop\ITR> █
```

Q9.

PROGRAM:

```
session 9.py > ...
1  contacts = {}
2
3  name = input("Enter name: ")
4  phone = input("Enter phone number: ")
5  email = input("Enter email: ")
6
7  # Store information
8  contacts[name] = {
9      "phone": phone,
10     "email": email
11 }
12
13 # Search contact
14 search_name = input("Enter name to search: ")
15
16 result = contacts.get(search_name)
17
18 if result:
19     print(result)
20 else:
21     print("Contact not found")
22
23 # Display all contacts
24 for name, details in contacts.items():
25     print(name, ":", details)
```

OUTPUT:

```
ions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\la
Enter name: shruti
Enter phone number: 86685918
Enter email: shruti@gmail.com
Enter name to search: shruti
{'phone': '86685918', 'email': 'shruti@gmail.com'}
shruti : {'phone': '86685918', 'email': 'shruti@gmail.com'}
PS C:\Users\tanve\Desktop\ITR>
```

Q10.

PROGRAM:

```
session 9.py > ...
1  students = {}
2
3  while True:
4      print("\n1.Add Student")
5      print("2.Update Marks")
6      print("3.Search Student")
7      print("4.Display All Students")
8      print("5.Remove Student")
9      print("6.Exit")
10
11     choice = int(input("Enter choice: "))
12
13     if choice == 1:
14         roll = input("Enter Roll No: ")
15         name = input("Enter Name: ")
16         age = int(input("Enter Age: "))
17         marks = int(input("Enter Marks: "))
18
19         students[roll] = {
20             "name": name,
21             "age": age,
22             "marks": marks
23         }
24
25     elif choice == 2:
26         roll = input("Enter Roll No: ")
27         if roll in students:
28             new_marks = int(input("Enter new marks: "))
29             students[roll]["marks"] = new_marks
30         else:
31             print("Student not found")
32
33     elif choice == 3:
34         roll = input("Enter Roll No: ")
35         print(students.get(roll, "Student not found"))
36
37     elif choice == 4:
38         for roll, details in students.items():
39             print(roll, ":", details)
40
41     elif choice == 5:
42         roll = input("Enter Roll No: ")
43         removed = students.pop(roll, "Student not found")
44         print(removed)
45
46     elif choice == 6:
47         print("Program Ended")
48         break
49
50     else:
51         print("Invalid choice")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\AppData\Local\Microsoft\Windows\i
ons\ms-python.debugpy-2026.6.0-win32-x64\bundled\Python\Scripts'

1.Add Student
2.Update Marks
3.Search Student
4.Display All Students
5.Remove Student
6.Exit
Enter choice: 4
14 : {'name': 'shruti', 'age': 18, 'marks': 89}

1.Add Student
2.Update Marks
3.Search Student
4.Display All Students
5.Remove Student
6.Exit
Enter choice: 5
Enter Roll No: 55
Student not found

1.Add Student
2.Update Marks
3.Search Student
4.Display All Students
5.Remove Student
6.Exit
Enter choice: 6
Program Ended
PS C:\Users\tanve\Desktop\ITR> |
```